

Surface #2 (Research Program): A Polymer-Expansion Scaffold for 4D SU(3) Lattice Yang–Mills, With an Honest Accounting of the Open Analytic Gaps

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Abstract

This note states the polymer-expansion route to a convergent cluster expansion for 4D SU(3) lattice Yang–Mills at small coupling, as the intended consumer of the Peter–Weyl heat-kernel infrastructure established in Surface #1 (Installment A). It is **not** a proof of convergence: the central inequalities — the per-plaquette exponential activity bound, the Kotecky–Preiss criterion, absolute convergence of the polymer log-partition series, and area-law decay of Wilson loops — are all open in the formalization. The corresponding theorem statement `kotecky_preiss_criterion` exists in `Towers/Attempts/ClusterExpansion.lean` as a parked `sorry`, and the surrounding YM cluster-expansion file carries roughly 60 additional sorries. This document fixes the *scaffold* that those sorries will eventually close into, and records the gaps explicitly so referees can audit them.

1 Status declaration

What is formalized (Surface #1, Installment A):

- `PeterWeyl_Summable_SU3` — summability of the Peter–Weyl heat-kernel series for every $t > 0$.
- `Weyl_sum_explicit_SU3_real_le_Heat_kernel_envelope_real` — truncation $\leq$ genuine tsum envelope.
- `Heat_kernel_envelope_real_ge_one_of_pos`.

Each closes under axiom footprint `{propext, Classical.choice, Quot.sound}`.

What is not formalized (this surface): The Varadhan small- t asymptotic for $K(t)$, the per-plaquette exponential activity bound, the Kotecky–Preiss criterion as a true theorem, the cluster expansion $\log Z_\Lambda$, the Ursell-function identity, and exponential decay of Wilson-loop expectations. The shipped statement of `kotecky_preiss_criterion` is a `sorry` in the project’s `Attempts/` directory; the Yang–Mills tower in `docs/ROADMAP.md` §2 remains *Status: Open*.

2 Definitions (informal scaffold)

Definition 2.1 (Polymer). *A polymer γ is a finite, geometrically connected set of plaquettes in the 4D lattice $\mathbb{Z}^4$. Let $\mathcal{P}$ denote the set of polymers.* Not yet formalized as a Lean structure; the project ships a placeholder `Polymer` type in `Towers/Attempts/ClusterExpansion.lean`.

Definition 2.2 (Polymer activity, target shape). *For inverse coupling $\beta > 0$ and Peter–Weyl truncation $N \in \mathbb{N}$,*

$$w(\gamma) := \prod_{p \in \gamma} \text{plaquette_activity_pw}(\beta, N, p).$$

The product structure and combinatorial dependence on γ are not formalized; only the per-plaquette activity placeholder exists.

Definition 2.3 (Kotecky–Preiss condition, target shape). *The expansion is said to satisfy KP at β if there exists $\mu : \mathcal{P} \rightarrow [0, \infty)$ such that for every $\gamma \in \mathcal{P}$,*

$$\sum_{\gamma' : \gamma' \cap \gamma \neq \emptyset} w(\gamma') e^{\mu(\gamma')} \leq \mu(\gamma).$$

3 Target theorems (currently sorry)

Conjecture (Per-plaquette exponential activity, target of `exists_c_per_plaquette_pw`). *There exist $C, c > 0$ and $t_0 > 0$ such that for every $\beta > 1/t_0$ and every plaquette p ,*

$$\text{plaquette_activity_pw}(\beta, N, p) \leq C \beta^{-4} e^{-c/\beta}.$$

Open Gap. *Conjecture ?? is the direct consumer of the Varadhan small- t asymptotic of Surface #1 (project task #156). Until that asymptotic is formalized, no quantitative per-plaquette decay rate is available; only the qualitative envelope bound $S_N(t) \leq K(t)$ of Surface #1, Installment A.*

Conjecture (KP criterion at small coupling, target of `kotecky_preiss_criterion`). *There exists $\beta_0 > 0$ such that for every $\beta \geq \beta_0$ the KP condition holds with $\mu(\gamma) = a|\gamma|$ for some $a > 0$ depending only on C, c from Conjecture ?? and on the 4D plaquette-graph branching constant.*

Open Gap. *The Lean identifier `kotecky_preiss_criterion` lives in `Towers/Attempts/ClusterExpansion.lean` (line ≈ 601) and is a **sorry**. The surrounding file carries roughly 60 further sorries; `Towers/YM/ClusterExpansion.lean` carries another ~ 60 . The combinatorial geometric-series argument (“sum over polymers intersecting γ is controlled by the branching constant raised to $|\gamma|$, times the activity bound”) is standard in the informal literature but requires formalizing (i) the polymer-graph branching count and (ii) the partial-sum estimate. Neither has been done in this project.*

Conjecture (Cluster expansion of $\log Z_\Lambda$). *For β as in Conjecture ??,*

$$\log Z_\Lambda(\beta) = \sum_X \Phi^T(X) \prod_{\gamma \in X} w(\gamma),$$

where X ranges over clusters and Φ^T is the Ursell function. The series converges absolutely.

Open Gap. Z_Λ , Φ^T , and the cluster sum are not formalized; the identity is the standard Kotecky–Preiss / Brydges–Federbush output, but a machine-checked derivation in this codebase does not exist.

Conjecture (Area-law decay of Wilson loops). *For β as above, every contractible Wilson loop W_C satisfies*

$$|\langle W_C \rangle| \leq K e^{-m \text{Area}(C)}$$

for some $K, m > 0$.

Open Gap. *Wilson loops are not defined in the codebase. The standard derivation (area law from convergent cluster expansion at strong coupling on the lattice) is conditional on Conjectures ??–??; it is also lattice-only and does not, by itself, address the continuum-limit mass-gap statement of the Clay problem. The continuum limit $a \rightarrow 0$ is its own surface and is not addressed here.*

4 Honest scope

This document does not claim to prove or formalize any of the conjectures above. It fixes the *shape* of the polymer-expansion argument that the project intends to close, lists the open gaps by Lean identifier, and is consistent with the project’s locked rule that the Yang–Mills tower remains *Status: Open* until the Lean spine actually closes the named theorem with axiom debt empty.

Cross-references

Lean identifiers referenced:

- `plaquette_activity_pw`, `exists_c_per_plaquette_pw`, `polymer_activity_bound_real_pw`, `kotecky_preiss_criterion` — in `Towers/YM/ClusterExpansion.lean` and `Towers/Attempts/ClusterExpansion.lean`.
- `Heat_kernel_envelope_real`, `Weyl_sum_explicit_SU3_real_le_Heat_kernel_envelope_real`, `Heat_kernel_envelope_real` — in `Towers/YM/PeterWeylHeat.lean`.
- `PeterWeyl_Summable_SU3` — in `Towers/YM/PeterWeyl.lean`.

Project task tracker:

- Task #154 — Peter–Weyl summability (closed).
- Task #155 — truncation-to-envelope bridge (closed; this is the entirety of formalized Surface #1 progress).
- Task #156 — Varadhan small- t asymptotic (open, blocks Surface #2).
- Task #157 — tightened Casimir / Weyl-dim envelopes (open, tech-debt; would shorten Surface #1 proofs).

Genesis seal of the project ledger `data/hits.txt` at the time of writing: `eecbcd9a540aa7a2c90edd23827c73e4d1bb5af641d3b`